



NITTE
(Deemed to be University)

**NITTE MEENAKSHI
INSTITUTE OF TECHNOLOGY**

Department of computer science and engineering

MAJOR PROJECT WORK- II (21CSP82)

Project Title: Edusphere: AI and Web Integrated Platform for Cognitive learning
Project Group No: 21CS44

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Background

- The education sector has struggled for years with inefficiencies in resource allocation, attendance management, communication, and the lack of personalized learning experiences.
- In the age of digitization, Artificial Intelligence (AI) provides promising solutions to these challenges.
- The platform addresses key pain points such as attendance fraud detection, personalized learning pathways, automated test creation and grading, and efficient resource sharing, all of which contribute to a holistic academic environment.

Research motivation and Problem Statement

- Attendance tracking often relies on outdated manual methods, leading to inaccuracies and time wastage. Students frequently lack access to personalized learning resources, limiting their ability to progress at their own pace. Communication between teachers and students remains fragmented, reducing opportunities for timely support.
- To address these issues, there is a need for a unified, AI-driven platform that can streamline educational processes.

Research Objectives

- To analyze and address inefficiencies within traditional educational systems, specifically focusing on manual operations such as attendance tracking.
- To investigate the integration of AI-based solutions in education, assessing their efficiency in optimizing processes and improving learning outcomes.
- To study the influence of AI applications on student engagement and teacher productivity, particularly in the context of facial recognition technologies.
- To review and synthesize existing research to identify both advantages and challenges of adopting AI-driven educational technologies.
- To propose potential avenues for future exploration and suggest strategies for developing scalable and user-centric AI innovations in education.

Literature Survey

- **Chatbots & Virtual Learning Systems**
 - Models: Random Forest, Ed.dy, Telegram Chatbot
 - Chatbot: Used for managing student queries and classroom interaction with an F-measure score of 0.870.
 - Ed.dy: Assisted virtual learning through organizational tools like reminders and polls.
 - Telegram Chatbot: Organized queries and tracked grades for programming courses, improving student engagement.

Literature Survey

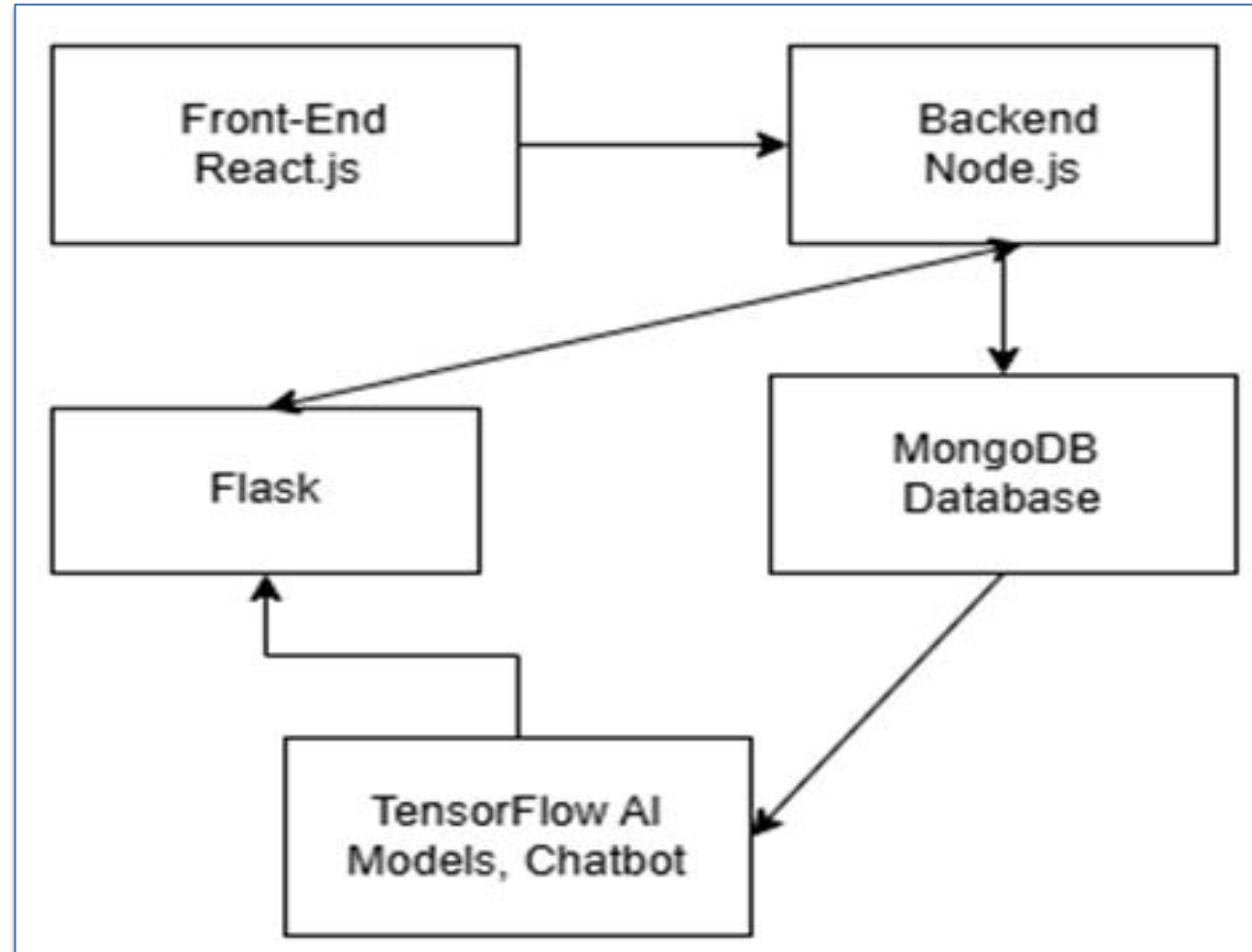
- **Deepfake Detection**
 - ➔ Models: Dense CNN, Large Language Models (LLMs), Meta-learning, Xception
 - ➔ Dense CNN + LLM: Used to detect deepfakes in multimedia content with 96.5% accuracy.
 - ➔ Meta-learning approach: Enhanced detection accuracy of deepfakes across datasets (FF++, DFDC, CelebDF-V2), outperforming Xception.

- **Data Scraping & Information Retrieval**
 - ➔ Tools: BeautifulSoup, Selenium, Scrapy
 - ➔ Web Scraping: Automated extraction of product data to identify illegal listings. Scrapy was used for efficient data extraction through machine learning integration for dynamic content handling.

Literature Survey

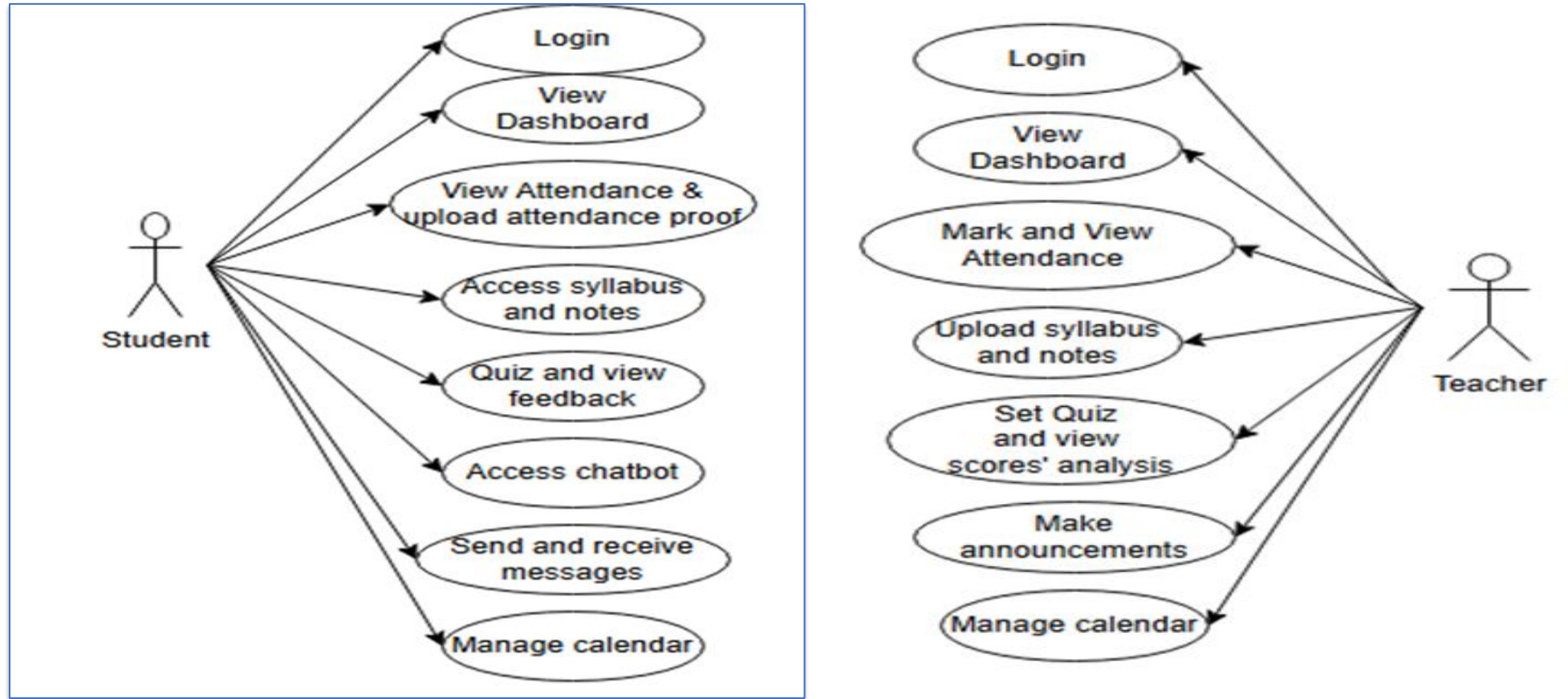
- **Facial Recognition & Attendance Systems**
 - Models: SVM, PCA, CNN, Haar Cascade Classifier, LBPH, HOG, KNN, Eigenface
 - Automated Attendance System utilized facial recognition for automated and efficient attendance marking. Vision-Face System enhanced attendance accuracy despite poor lighting and proxy attendance.
 - Hybrid Model: Achieved 94.32% accuracy in attendance marking using multiple facial recognition techniques.
 - Face Recognition System: Integrated algorithms for 100% attendance accuracy.

ARCHITECTURAL DESIGN



Design and Implementation

Usecase diagram



Design and Implementation

Technology Stack:

Frontend: ReactJS

Backend: Node.js, Flask

Database: MongoDB for data storage.

AI/ML Libraries: TensorFlow, OpenCV

Web Scraping Tools: BeautifulSoup

Cloud Services: AWS or Azure for deployment and storage

Design and Implementation

Attendance Management System

❖ Dataset:

The dataset consists of a single root folder containing 5 subfolders. Each subfolder is named after a person (e.g., "angelina_jolie", "bradley_cooper", etc.) and contains 1 image of that person.

❖ Preprocessing:

- Background Removal (using GrabCut)
- Face Detection
- Normalization
- HOG Feature Extraction
- Augmentation

Design and Implementation

Attendance Management System

- **Support Vector Machine (SVM):**

Feature Extraction: The SVM model uses HOG features as input to classify the images. The HOG features are effective for capturing spatial and textural information about faces.

Training: The SVM classifier was trained with a linear kernel and probability estimates enabled.

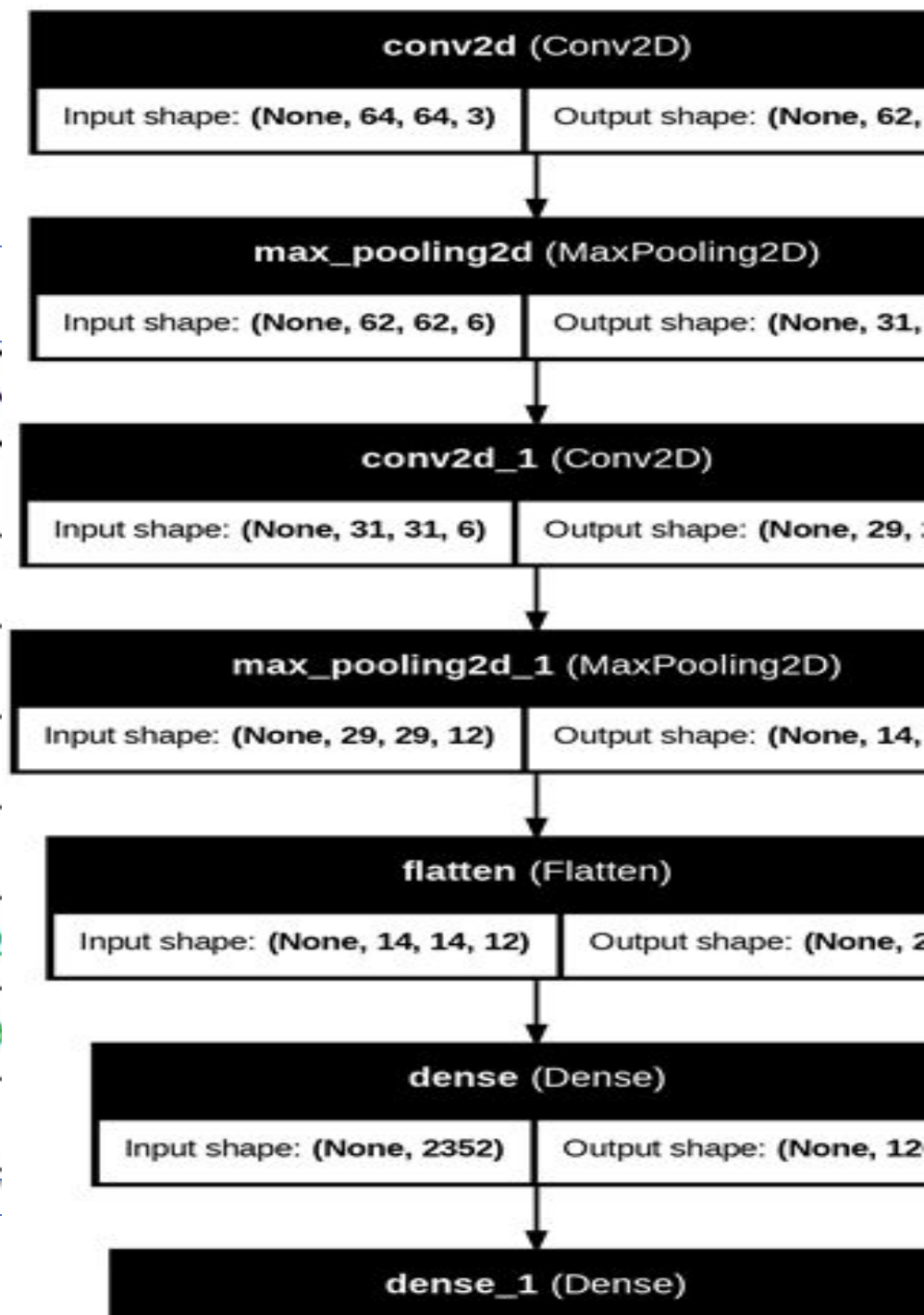
- **K-Nearest Neighbors (KNN):**

Feature Extraction: Like SVM, the KNN model also uses HOG features. KNN works by classifying a new image based on the majority label of the nearest neighbors (images with similar features).

Training: The KNN model was trained using Euclidean distance as the metric for similarity and uses 5 nearest neighbors.

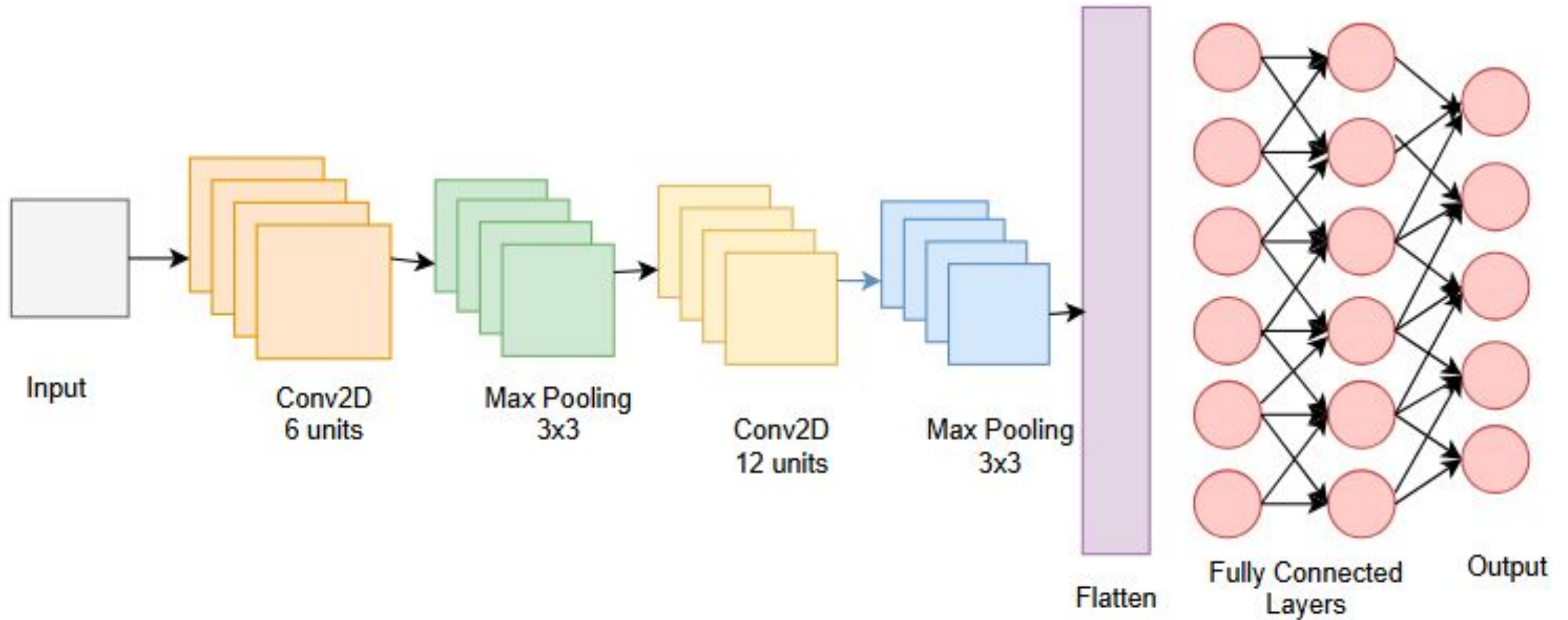
Convolutional Neural Network (CNN)

Layer (type)	Output Shape	Par
conv2d (Conv2D)	(None, 62, 62, 6)	
max_pooling2d (MaxPooling2D)	(None, 31, 31, 6)	
conv2d_1 (Conv2D)	(None, 29, 29, 12)	
max_pooling2d_1 (MaxPooling2D)	(None, 14, 14, 12)	
flatten (Flatten)	(None, 2352)	
dense (Dense)	(None, 120)	282
dense_1 (Dense)	(None, 84)	10
dense_2 (Dense)	(None, 5)	



Design and Implementation

Convolutional Neural Network (CNN)



Design and Implementation

Attendance Management System: Image Editing Detection

Dataset:

The dataset used in the code is the CASIA2 dataset, which is available on Kaggle. The dataset is structured into two subfolders. Au (Authentic) folder contains authentic (real) images. Tp (Tampered) folder contains tampered (fake) images.

Preprocessing:

- Balancing the Dataset
- ELA
- Edge Detection
- Image Resizing

Design and Implementation

Attendance Management System: Image Editing Detection

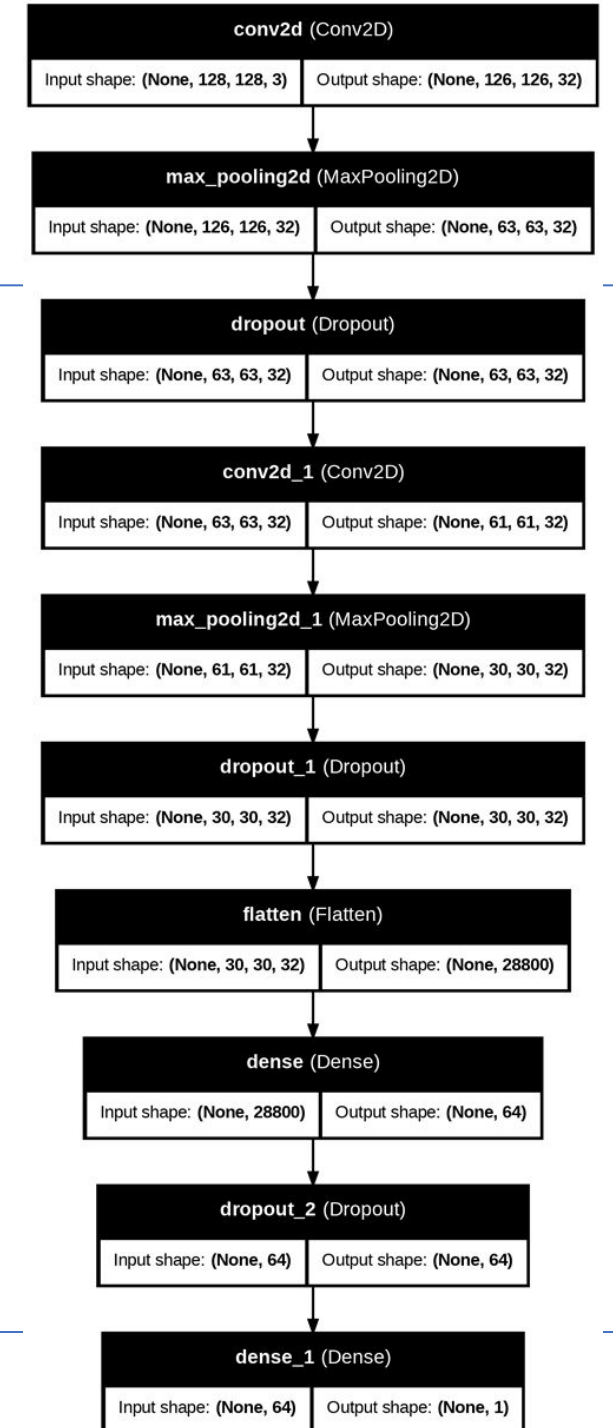
- **Random Forest Classifier:**

Architecture: A traditional machine learning model, the Random Forest Classifier uses a collection of decision trees to make predictions. It is trained on features extracted from the images, such as Error-Level Analysis (ELA) and edge features.

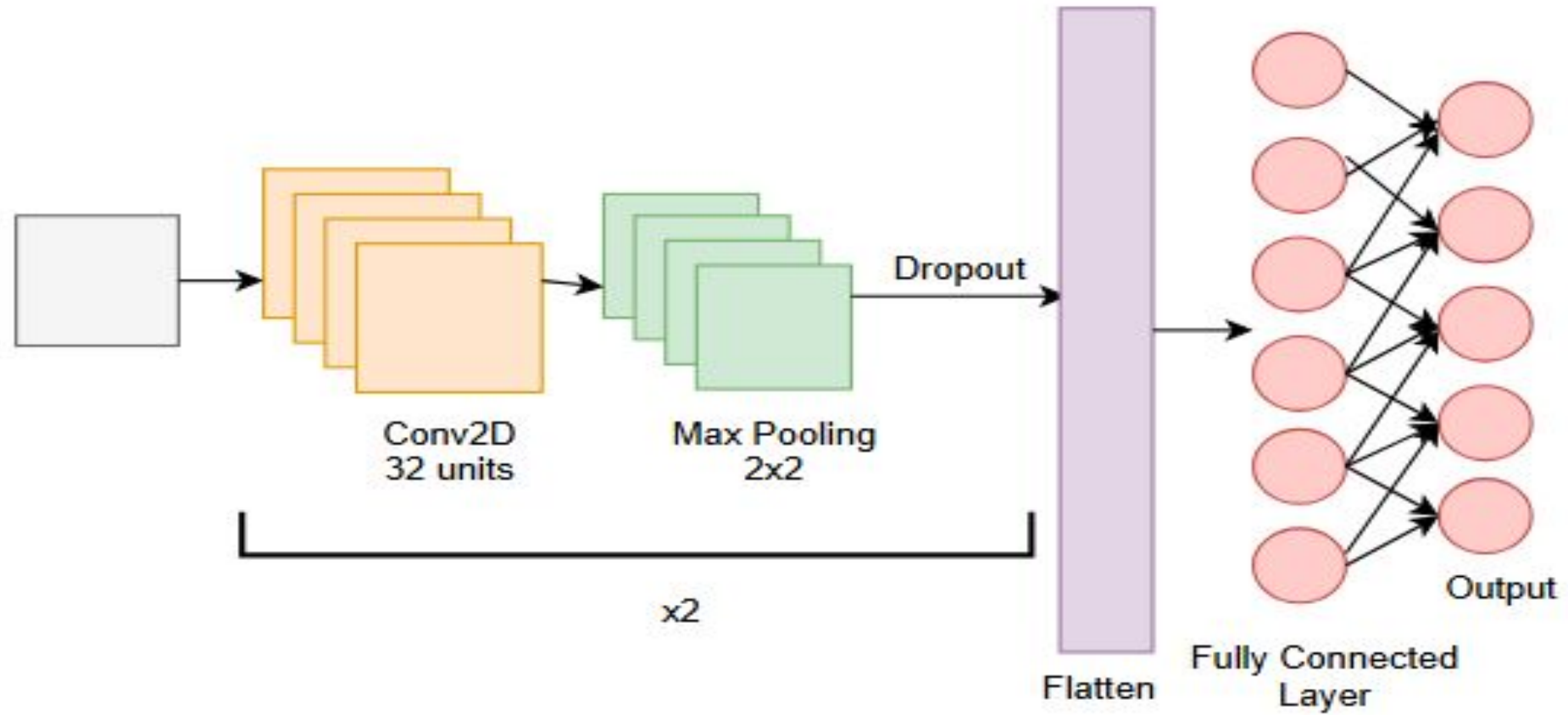
Training: The Random Forest model is trained on the extracted features (ELA and edge intensity) rather than raw pixel values. Random Forests are typically used for classification tasks, and in this case, it is used to differentiate between authentic and tampered images based on hand-crafted features.

Design and Implementation

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 126, 126, 32)	896
max_pooling2d (MaxPooling2D)	(None, 63, 63, 32)	0
dropout (Dropout)	(None, 63, 63, 32)	0
conv2d_1 (Conv2D)	(None, 61, 61, 32)	9,248
max_pooling2d_1 (MaxPooling2D)	(None, 30, 30, 32)	0
dropout_1 (Dropout)	(None, 30, 30, 32)	0
flatten (Flatten)	(None, 28800)	0
dense (Dense)	(None, 64)	1,843,264
dropout_2 (Dropout)	(None, 64)	0
dense_1 (Dense)	(None, 1)	65



Design and Implementation CNN Fake Image Detection

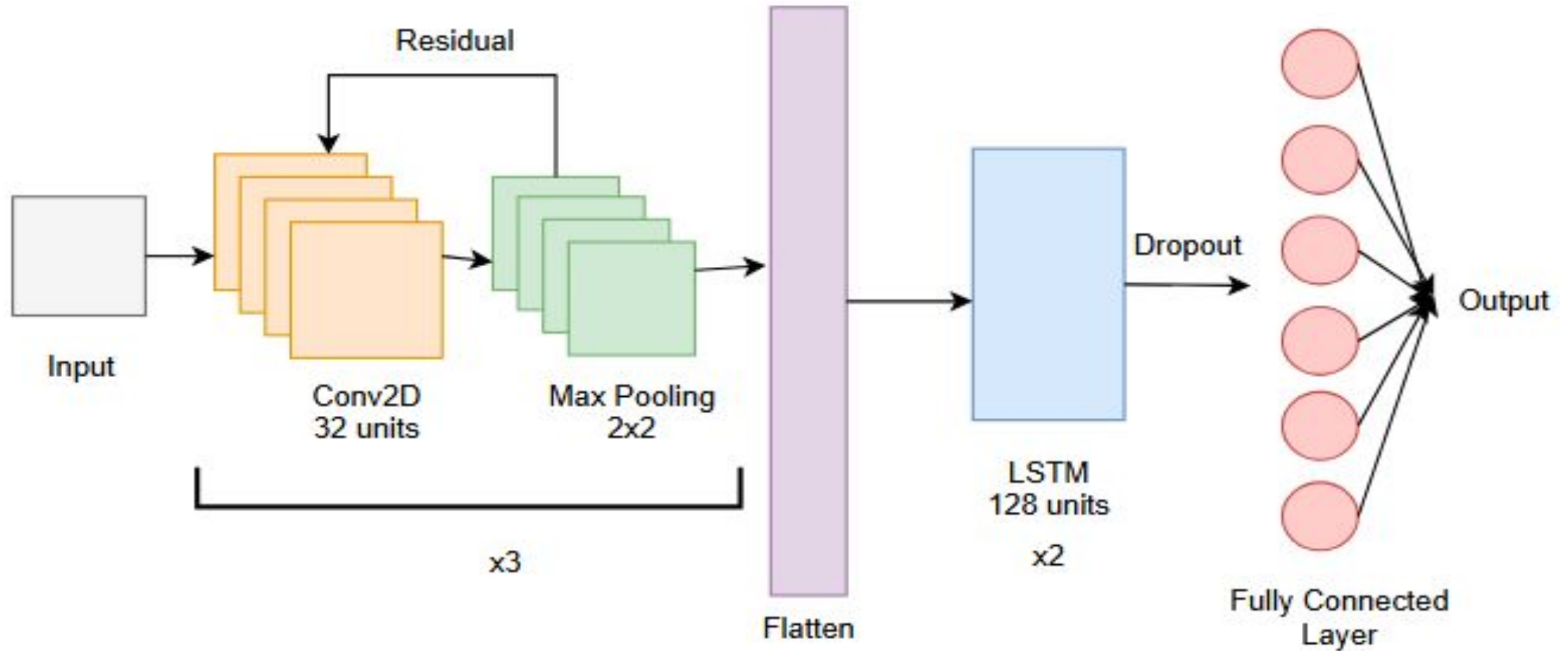




Design and Implementation Hybrid(CNN+LSTM(Long Short-term Memory))

Layer (type)	Output Shape	Param #	Connected to
input_layer (InputLayer)	(None, 128, 128, 3)	0	-
conv2d (Conv2D)	(None, 128, 128, 32)	896	input_layer[0][0]
conv2d_1 (Conv2D)	(None, 128, 128, 32)	9,248	conv2d[0][0]
add (Add)	(None, 128, 128, 32)	0	conv2d_1[0][0], conv2d[0][0]
max_pooling2d (MaxPooling2D)	(None, 64, 64, 32)	0	add[0][0]
conv2d_2 (Conv2D)	(None, 64, 64, 64)	18,496	max_pooling2d[0][0]
conv2d_3 (Conv2D)	(None, 64, 64, 64)	36,928	conv2d_2[0][0]
add_1 (Add)	(None, 64, 64, 64)	0	conv2d_3[0][0], conv2d_2[0][0]
max_pooling2d_1 (MaxPooling2D)	(None, 32, 32, 64)	0	add_1[0][0]
conv2d_4 (Conv2D)	(None, 32, 32, 128)	73,856	max_pooling2d_1[0][0]
conv2d_5 (Conv2D)	(None, 32, 32, 128)	147,584	conv2d_4[0][0]
add_2 (Add)	(None, 32, 32, 128)	0	conv2d_5[0][0], conv2d_4[0][0]
max_pooling2d_2 (MaxPooling2D)	(None, 16, 16, 128)	0	add_2[0][0]
flatten_2 (Flatten)	(None, 32768)	0	max_pooling2d_2[0][0]
reshape_6 (Reshape)	(None, 1, 32768)	0	flatten_2[0][0]
lstm_4 (LSTM)	(None, 1, 128)	16,843,264	reshape_6[0][0]
dropout_4 (Dropout)	(None, 1, 128)	0	lstm_4[0][0]
reshape_7 (Reshape)	(None, 1, 128)	0	dropout_4[0][0]
lstm_5 (LSTM)	(None, 128)	131,584	reshape_7[0][0]
dropout_5 (Dropout)	(None, 128)	0	lstm_5[0][0]
dense (Dense)	(None, 1)	129	dropout_5[0][0]

Design and Implementation Hybrid(CNN+LSTM(Long Short-term Memory))



Design and Implementation

Web Scraping

Procedure:

The web scraping module aggregates educational resources, interview preparation materials, and relevant links for students. Python's BeautifulSoup library was employed to extract content from static websites.

Data Filtering: Extracted content was filtered to retain only links and materials relevant to predefined educational topics.

Content Display: Collected resources were categorized and presented in an organized manner for user access.

Tools and Libraries:

Beautiful Soup: For parsing HTML content.

Design and Implementation

Edusphere AI

Procedure:

Using the requests library in Python, the chatbot sends the input to the Hugging Face API by making an HTTP POST request. Each model has a unique endpoint, specified by the API URL. The Hugging Face models process the input. The response from the Hugging Face API is then returned as a JSON object, which contains the output from the model which is displayed to the user.

Features:

- Text Summarization
- Question-Answering

Design and Implementation

LMS Frontend and Backend Development

- Use React components like calendars
- Style using Material-UI or custom CSS frameworks.
- State Management: Use of hooks to manage the change of state..
- APIs and Data Binding: Fetch data dynamically from the backend using axios or fetch. Display progress and analytics with tools like Chart.js.
- User Authentication: Use custom backend logic for secure login/signup flows. Implement role-based access controls.
- Backend Integration: Create APIs to fetch events and tasks.

Results

Comparative analysis of Face Recognition Models

Algorithm	Accuracy	Precision	Recall	F1-Score	Test Loss
CNN	99.37%	-	-	-	0.0222
SVM	100.00%	1.00	1.00	1.00	-
KNN	100.00%	1.00	1.00	1.00	-

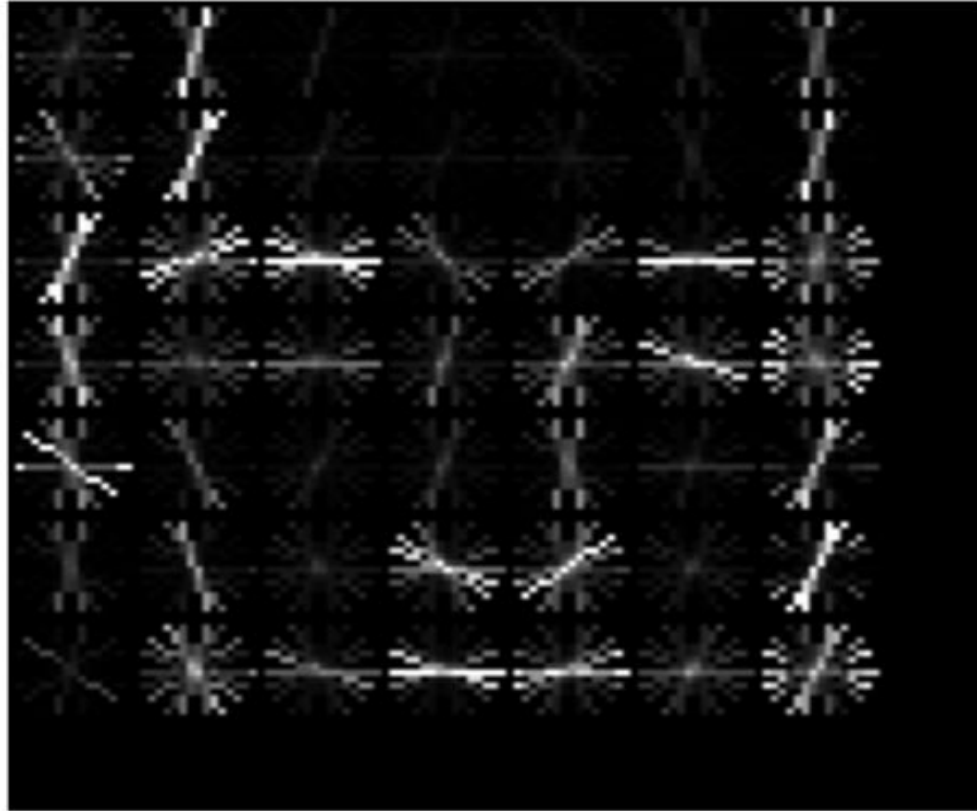
- SVM (Support Vector Machine) and KNN (K-Nearest Neighbors) both achieved perfect accuracy (100%).
- The CNN (Convolutional Neural Network) also performs exceptionally well with an accuracy of 99.37%.

Results

Face Recognition Output

Preprocessed input image shape: (1, 64, 64, 3)

HOG Visualization: pr.jpg



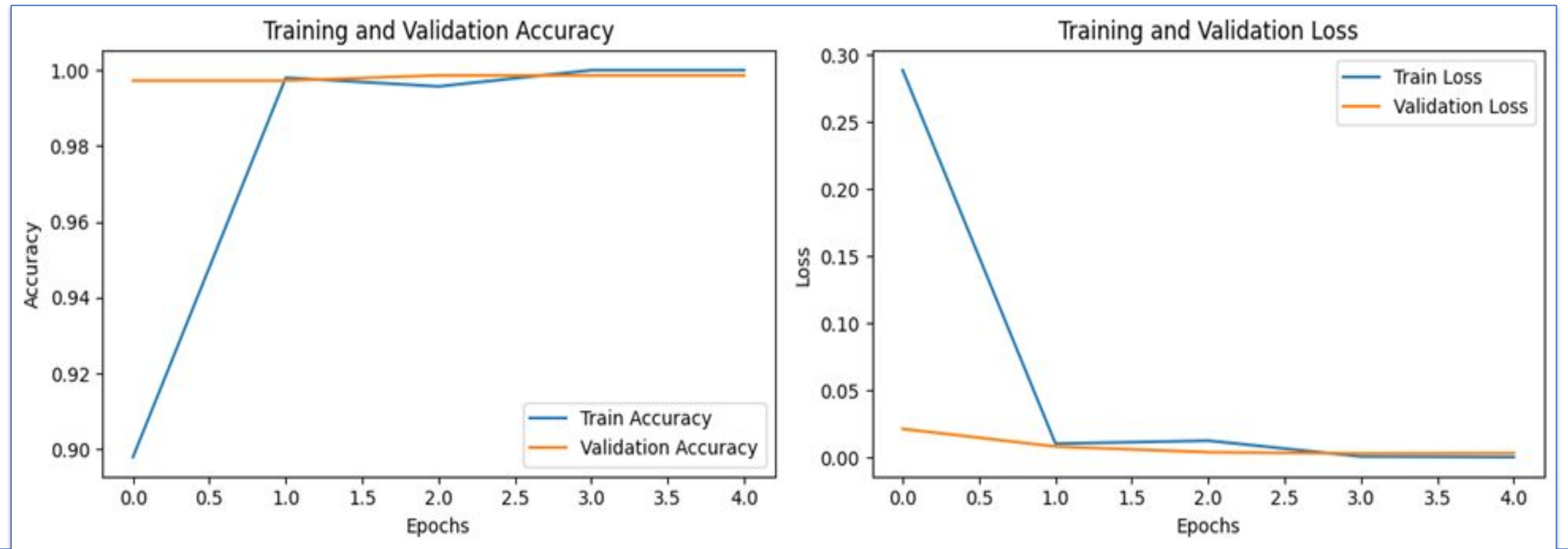
1/1 ————— 0s 75ms/step

Predictions: $[[0.0000000e+00 \ 3.1351868e-15 \ 0.0000000e+00 \ 1.0000000e+00 \ 0.0000000e+00]]$

Predicted person: paul_rudd

Results

Training and Validation Accuracy and Loss graph for CNN Face Recognition



Results

Image Editing Classification

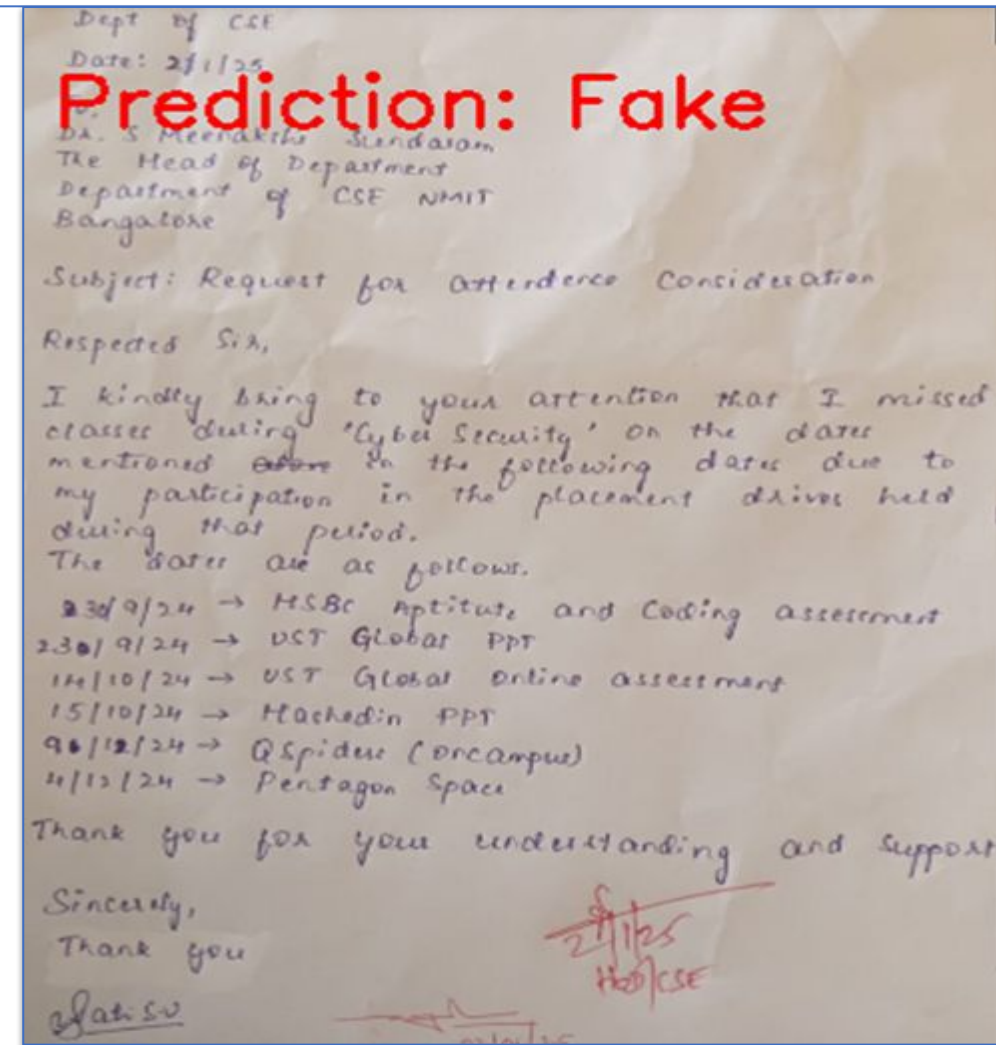
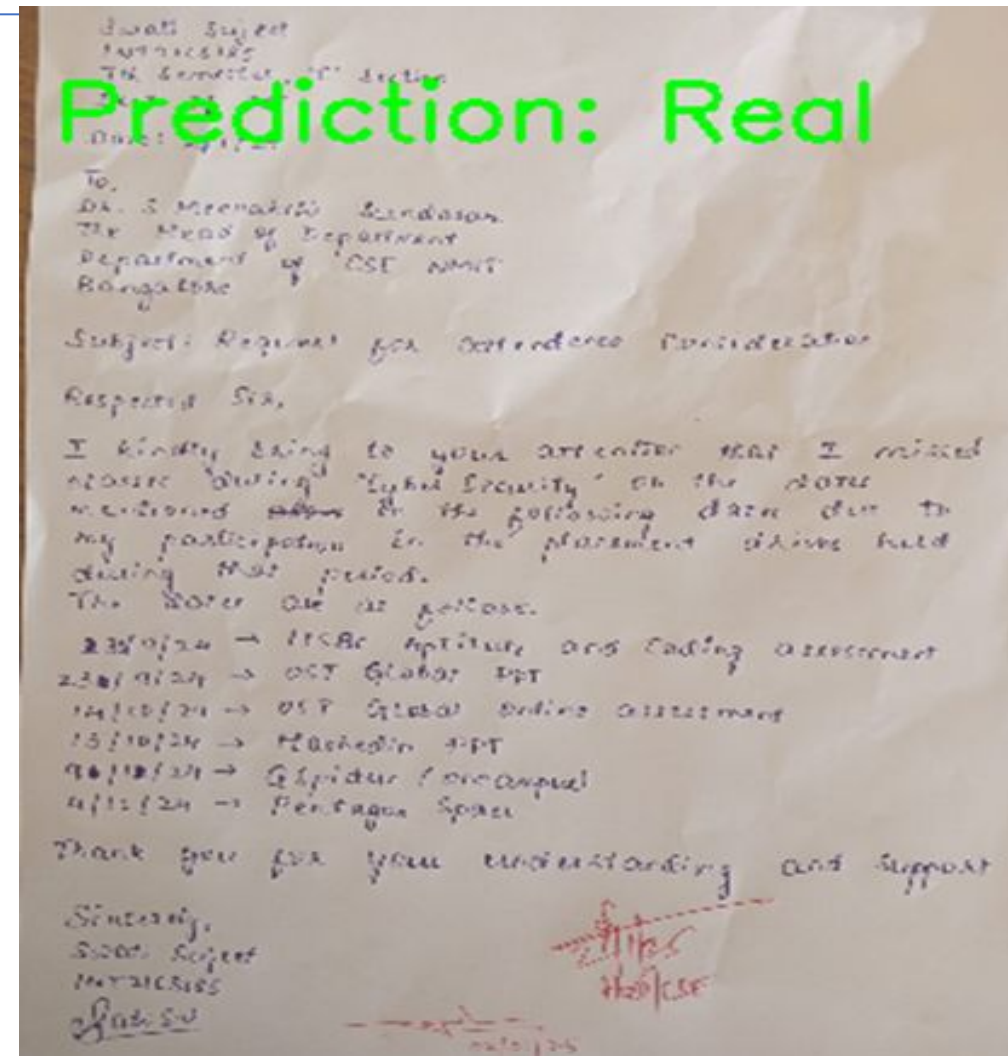
Comparative analysis of Image Editing Classification Models

Algorithm	Accuracy	Precision	Recall	F1-Score	Test Loss
CNN	88.81%	-	-	-	0.3168
CNN+LSTM	89.12%	-	-	-	0.2385
Random Forest	77.25%	0.77	0.77	0.77	-

- CNN and CNN+LSTM performed similarly with high accuracy, but CNN+LSTM slightly edged out CNN with an accuracy of 89.12%.
- Random Forest performed considerably lower, with an accuracy of 77.25%.

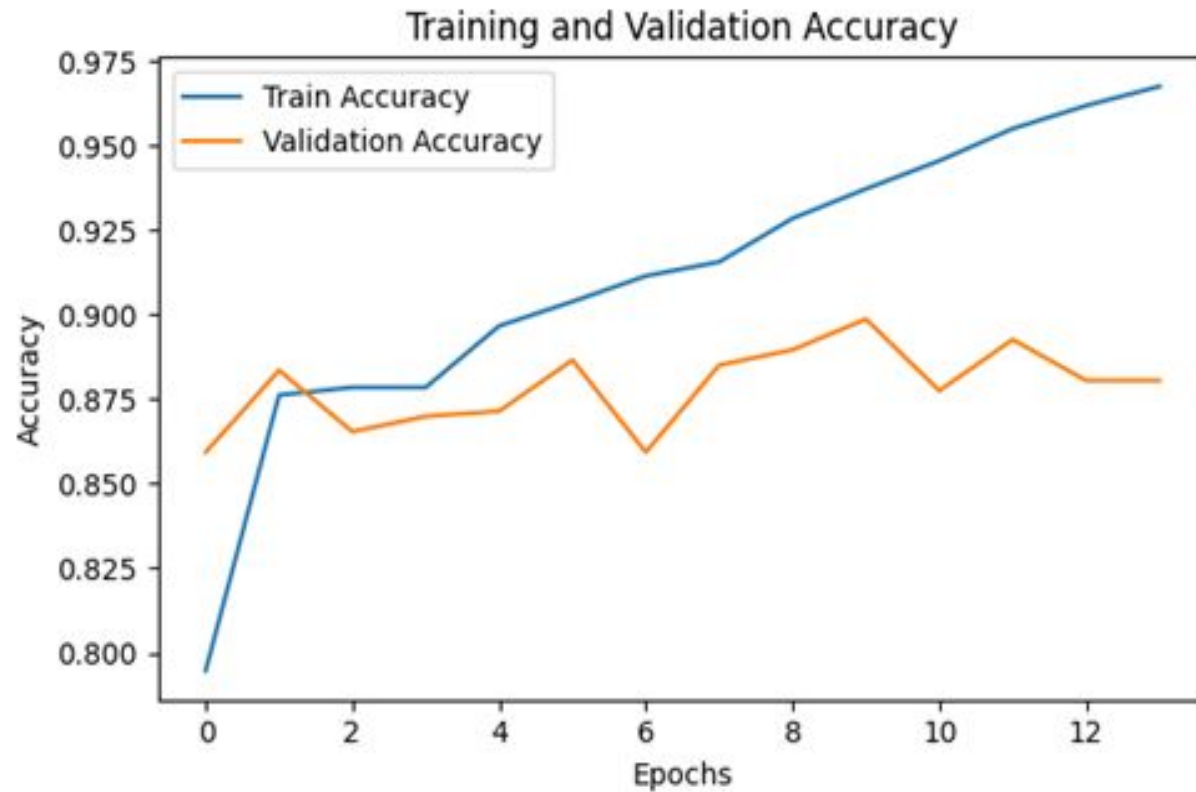
Results

Image Editing Classification : Real(left) Fake(right)



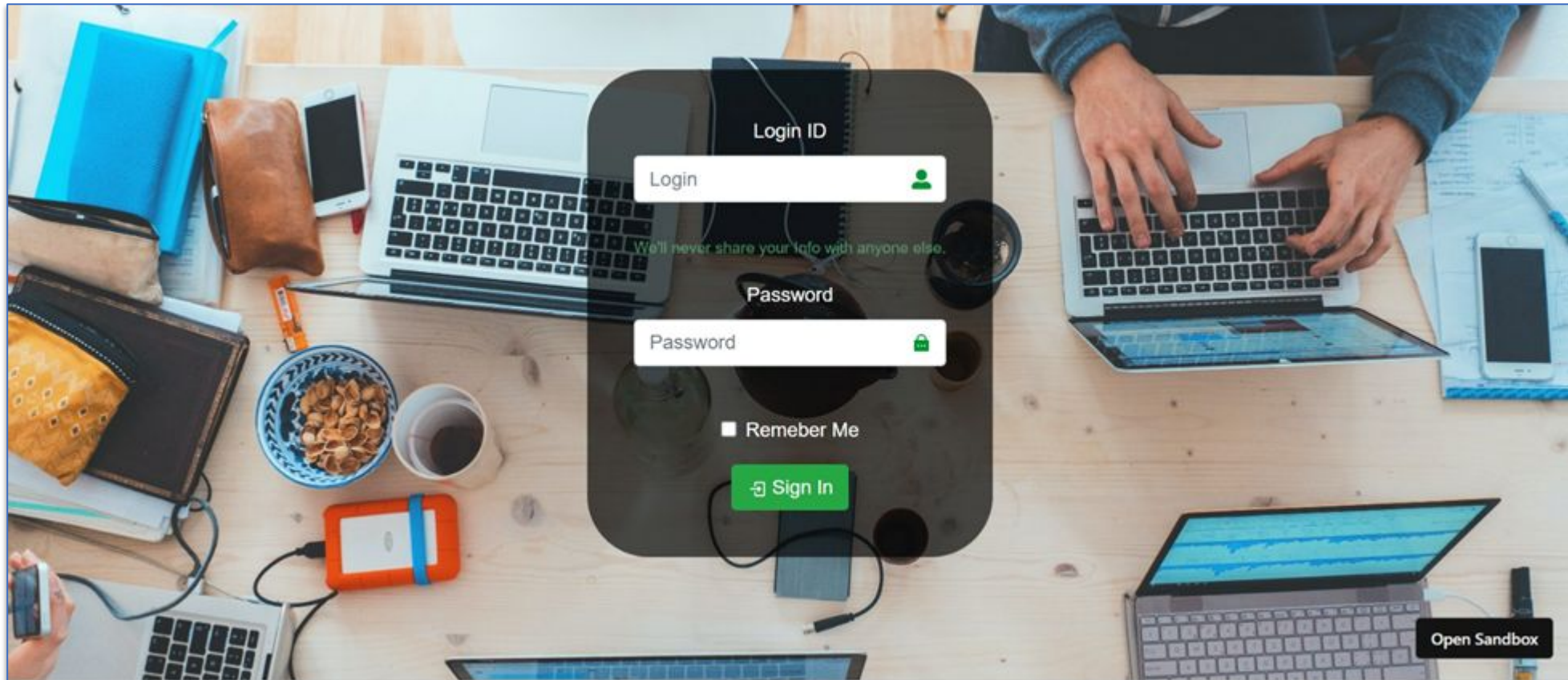
Results

Training and Validation Accuracy and Loss graph for CNN Image Editing



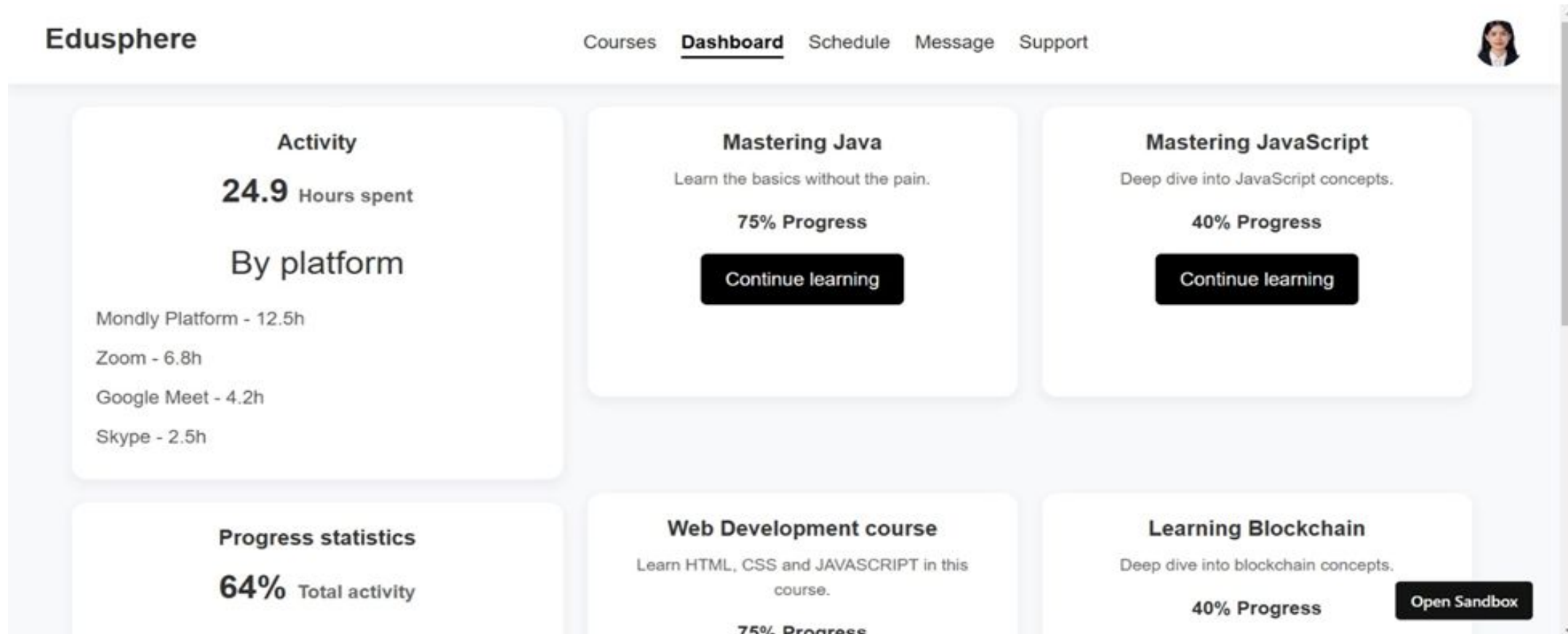
Results

Login page



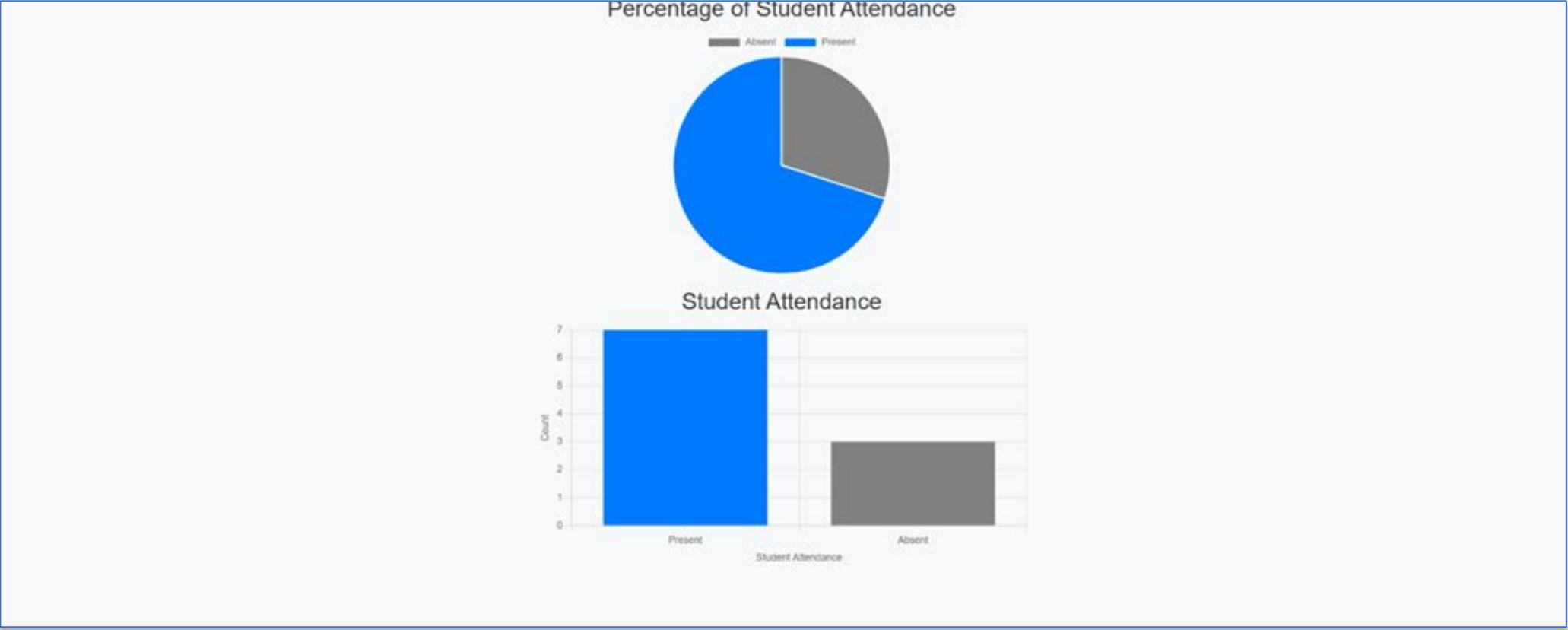
Results

Dashboard - Courses, activity, profile, top bar

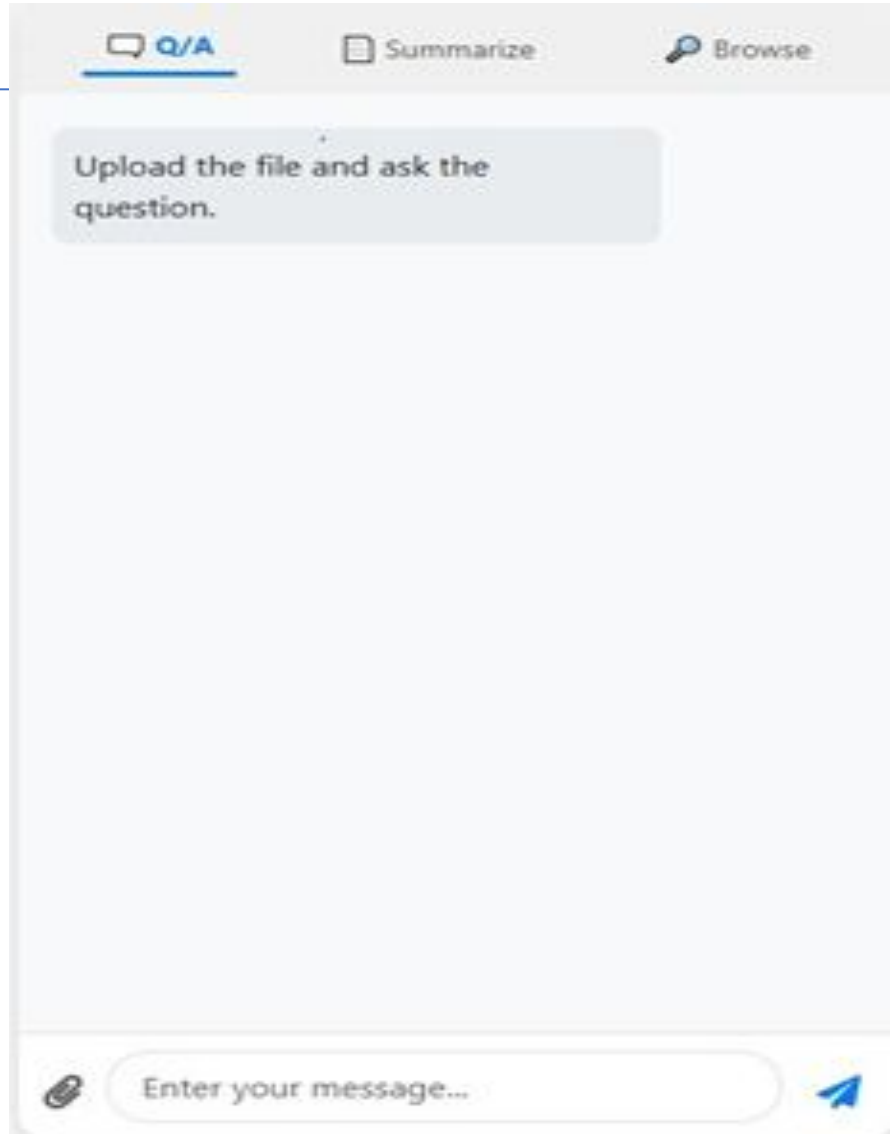


Results

Attendance Page



Edusphere AI Interface



Publications status with proof

- Research Paper has been accepted for the 2025 Second International Conference on Advances in Modern Age Technologies for Health and Engineering Science (AMATHE 2025) to be held at PES Institute of Technology and Management (PESITM), Shivamogga, Karnataka, India during 24 – 25, April 2025.
- Proof of Acceptance:
https://drive.google.com/file/d/1aNNrvjzf3dsuhhz7kGeJmV7qSMzhAggv/view?usp=drive_link

Conclusion & Future Work

- By developing an integrated platform that combines AI-driven attendance systems, personalized learning tools, and automated evaluation mechanisms, the project provides a scalable and efficient solution for students and educators.
- Through the use of advanced AI models for fraud detection, resource recommendations, and personalized feedback, the proposed system simplifies administrative tasks, enhances the learning experience, and fosters better collaboration between students and teachers.

The platform can be enhanced by integrating real-time analytics and adaptive learning paths tailored to individual student performance.

References

- [1] J. Cheng, "Design of Intelligent Autonomous Learning Management System Based on Artificial Intelligence Technology," 2022 International Conference on Education, Network and Information Technology (ICENIT), Liverpool, United Kingdom, 2022, pp. 106-110, doi: 10.1109/ICENIT57306.2022.00030.
- [2] S. Dutta, S. Ranjan, S. Mishra, V. Sharma, P. Hewage and C. Iwendi, "Enhancing Educational Adaptability: A Review and Analysis of AI Driven Adaptive Learning Platforms," 2024 4th International Conference on Innovative Practices in Technology and Management (ICIPTM), Noida, India, 2024, pp. 1-5, doi: 10.1109/ICIPTM59628.2024.10563448.
- [3] S. K. Vishnumolakala, S. C. C, N. P. Subheesh, P. Kumar and R. Kumar, "AI-Based Research Companion (ARC): An Innovative Tool for Fostering Research Activities in Undergraduate Engineering Education," 2024 IEEE Global Engineering Education Conference (EDUCON), Kos Island, Greece, 2024, pp. 1-5, doi: 10.1109/EDUCON60312.2024.10578646.
- [4] S. Zharmukhanbetov and C. P. Singh, "Enhancing Flipped Classroom Engagement and Personalized Learning Through AI-Powered Adaptive Content Delivery," 2023 3rd International Conference on Technological Advancements in Computational Sciences (ICTACS), Tashkent, Uzbekistan, 2023, pp. 1411-1416, doi: 10.1109/ICTACS59847.2023.10389841.
- [5] Y. Du, W. Song, Y. Coa, X. Chen and G. Huang, "Design of Personalized AI Examination System Based on Reinforcement Learning," 2024 6th International Conference on Computer Science and Technologies in Education (CSTE), Xi'an, China, 2024, pp. 119-124, doi: 10.1109/CSTE62025.2024.00029.

References

- [6] B. Pardamean, T. Suparyanto, T. W. Cenggoro, D. Sudigyo, A. Anugrahana and I. Anugraheni, "Model of Learning Management System Based on Artificial Intelligence in Team-Based Learning Framework," 2021 International Conference on Information Management and Technology (ICIMTech), Jakarta, Indonesia, 2021, pp. 37-42, doi: 10.1109/ICIMTech53080.2021.9535088.
- [7] J. Xiao and Q. Bai, "iTutor: Promoting AI Guided Knowledge Interaction in Online Learning," 2022 International Symposium on Educational Technology (ISET), Hong Kong, Hong Kong, 2022, pp. 253-257, doi: 10.1109/ISET55194.2022.00061.
- [8] J. Harikrishnan, A. Sudarsan, A. Sadashiv and R. A. S. Ajai, "Vision Face Recognition Attendance Monitoring System for Surveillance using Deep Learning Technology and Computer Vision," 2019 International Conference on Vision Towards Emerging Trends in Communication and Networking (ViTECoN), Vellore, India, 2019, pp. 1-5, doi: 10.1109/ViTECoN.2019.8899418.
- [9] H. Kesharwani, T. Mallick, A. Gupta and G. Raj, "Automated Attendance System Using Computer Vision," 2022 Second International Conference on Computer Science, Engineering and Applications (ICCSEA), Gunupur, India, 2022, pp. 1-5, doi:

*THANK
YOU !!!*